

Kuwait University

2021/2022

Spring course



Calculus B

Second

Sections : 11.6 – 11.8

11.6 Absolute Convergence and the Ratio and Root Tests

11.8 Power Series.

(12)



11.6 Ratio and Root tests



Ratio Test

شكله

عند وجود

a^n

عند وجود

factorial !

خطوات الحل

1) $a_n \rightarrow a_{n+1}$

2) $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$

< 1



Convergent

> 1



Divergent

$= 1$



فشل الاختبار

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}$$

$$a_n = \frac{(2n)!}{(n!)^2} \rightarrow a_{n+1} = \frac{(2(n+1))!}{((n+1)!)^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(2(n+1))!}{((n+1)!)^2} \cdot \frac{(n!)^2}{(2n)!} \right|$$

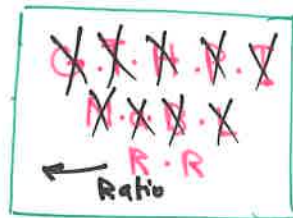
$$\lim_{n \rightarrow \infty} \left| \frac{(2n+2)!}{((n+1)!)^2} \cdot \frac{(n!)^2}{(2n)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(2n+2)(2n+1)(2n)!}{((n+1) \cdot n!)^2} \cdot \frac{(n!)^2}{(2n)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(2n+2)(2n+1)}{(n+1)^2} \cdot \frac{(n!)^2}{(n!)^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n(2+\frac{2}{n})n(2+\frac{1}{n})}{[n(1+\frac{1}{n})]^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^2 (2+\frac{2}{n})(2+\frac{1}{n})}{n^2 (1+\frac{1}{n})^2} \right| = \frac{(2+0)(2+0)}{1+0} = \frac{4}{1} = 4 > 1$$



يُظهِرُ التَّكْبِيرَ

∴ $\sum_{n=1}^{\infty} a_n$ is Divergent ((Ratio Test)) R.T

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sqrt{n!}}{(n)^n}$$

$$a_n = \frac{\sqrt{n!}}{(n)^n} \rightarrow a_{n+1} = \frac{\sqrt{(n+1)!}}{(n+1)^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

~~G.T.~~ ~~A.P.~~ ~~X~~
~~N.A.B.~~ ~~X~~
 R.R

$$\lim_{n \rightarrow \infty} \left| \frac{\sqrt{(n+1)!}}{(n+1)^{n+1}} \cdot \frac{n^n}{\sqrt{n!}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\sqrt{(n+1) n!}}{(n+1)^n \cdot (n+1)} \cdot \frac{n^n}{\sqrt{n!}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\sqrt{n+1} \sqrt{n!}}{(n+1)^n (n+1)} \cdot \frac{n^n}{\sqrt{n!}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\sqrt{n+1}}{n+1} \cdot \frac{n^n}{(n+1)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^{\frac{1}{2}}}{n+1} \cdot \left(\frac{n}{n+1} \right)^n \right|$$

$$\lim_{n \rightarrow \infty} \left| (n+1)^{\frac{1}{2}-1} \cdot \left(\frac{n+1-1}{n+1} \right)^n \right|$$

$$\lim_{n \rightarrow \infty} \left| (n+1)^{-\frac{1}{2}} \cdot \left(\frac{n+1}{n+1} + \frac{-1}{n+1} \right)^n \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{1}{\sqrt{n+1}} \cdot \left(1 + \frac{-1}{n+1} \right)^n \right| = \left| \frac{1}{\infty} \cdot e^{-1} \right| = 0 < 1$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent (Ratio Test)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{n}{4^n}$$

$$a_n = \frac{n}{4^n} \longrightarrow a_{n+1} = \frac{n+1}{4^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n+1}{4^{n+1}} \cdot \frac{4^n}{n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n+1}{4^n \cdot 4} \cdot \frac{4^n}{n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n(1 + \frac{1}{n})}{4 \cdot n} \right| = \left| \frac{1+0}{4} \right| = \frac{1}{4} < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{n}{4^n}$ is convergent (Ratio test)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{n^2}{\sqrt{e^n}} \rightarrow \sum a_n$$

$$a_n = \frac{n^2}{\sqrt{e^n}} \rightarrow a_{n+1} = \frac{(n+1)^2}{\sqrt{e^{n+1}}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{\sqrt{e^{n+1}}} \cdot \frac{\sqrt{e^n}}{n^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{\cancel{\sqrt{e^n}} \cdot e^{1/2}} \cdot \frac{\cancel{\sqrt{e^n}}}{n^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^2}{\sqrt{e}} \cdot \frac{1}{n^2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{[n(1 + \frac{1}{n})]^2}{\sqrt{e}} \cdot \frac{1}{n^2} \right| = \frac{1+0}{\sqrt{e}} \cdot 1 = \frac{1}{\sqrt{e}} < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is convergent (Ratio test)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{e^n \cos 2n}{(n+1)!}$$

$$-1 \leq \cos 2n \leq 1$$

$$* \frac{e^n}{(n+1)!}$$

~~C.T.A.P.X~~
~~N.X.B.L~~
R.R

$$-\frac{e^n}{(n+1)!} \leq \frac{e^n \cos 2n}{(n+1)!} \leq \frac{e^n}{(n+1)!} \quad \rightarrow b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{e^n}{(n+1)!}$$

$$b_n = \frac{e^n}{(n+1)!} \rightarrow b_{n+1} = \frac{e^{n+1}}{(n+2)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{e^{n+1}}{(n+2)!} \cdot \frac{(n+1)!}{e^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{e^n \cdot e^1}{(n+2)(n+1)!} \cdot \frac{(n+1)!}{e^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{e^1}{n+2} \right| = \left| \frac{e^1}{\infty} \right| = 0 < 1$$

$\therefore \sum_{n=1}^{\infty} b_n$ is convergent (Ratio Test)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)

Determine whether the series is Convergent or Divergent

$$\sum_{n=0}^{\infty} \frac{\cos(3n) (n!)^3 e^{3n}}{(3n)!} \rightarrow a_n$$

$-1 \leq \cos 3n \leq 1$

* $\frac{(n!)^3 e^{3n}}{(3n)!} \leq \frac{(n!)^3 e^{3n}}{(3n)!} \leq \frac{\cos 3n (n!)^3 e^{3n}}{(3n)!} \leq \frac{(n!)^3 e^{3n}}{(3n)!} \leftarrow b_n$

$$\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} \frac{(n!)^3 e^{3n}}{(3n)!}$$

$$b_n = \frac{(n!)^3 e^{3n}}{(3n)!} \rightarrow b_{n+1} = \frac{((n+1)!)^3 e^{3(n+1)}}{(3(n+1))!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{((n+1)!)^3 e^{3n+3}}{(3n+3)!} \cdot \frac{(3n)!}{(n!)^3 e^{3n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{[(n+1)(n!)]^3 e^{3n} e^3}{(3n+3)(3n+2)(3n+1)(3n)!} \cdot \frac{(3n)!}{(n!)^3 e^{3n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^3 (n!)^3 e^3}{(3n+3)(3n+2)(3n+1)} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{[n(1 + \frac{1}{n})]^3 e^3}{n(3 + \frac{3}{n}) n(3 + \frac{2}{n}) n(3 + \frac{1}{n})} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n^3 (1 + \frac{1}{n})^3 e^3}{n^3 (3 + \frac{3}{n})(3 + \frac{2}{n})(3 + \frac{1}{n})} \right| = \left| \frac{(1+0)^3 e^3}{(3+0)(3+0)(3+0)} \right| = \frac{e^3}{27} < 1$$

$\Rightarrow \therefore \sum_{n=0}^{\infty} a_n$ is convergent (B.C.T.)

Determine whether the series is convergent or divergent

$$\sum_{n=3}^{\infty} \frac{\ln n}{3^n} \cos^2 n$$

$\because \frac{\ln n}{3^n} > 0 \quad \forall n \in [3, \infty)$

$\ast \frac{\ln n}{3^n}$

$0 < \cos^2 n < 1$

$0 < \frac{\ln n}{3^n} \cdot \cos^2 n < \frac{\ln n}{3^n}$

b_n

$$\sum_{n=3}^{\infty} \frac{\ln n}{3^n}$$

$$b_n = \frac{\ln n}{3^n} \rightarrow b_{n+1} = \frac{\ln(n+1)}{3^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{b_{n+1}}{b_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\ln(n+1)}{3^{n+1}} \cdot \frac{3^n}{\ln n} \right|$$

$$\lim_{n \rightarrow \infty} \frac{\ln(n+1)}{\cancel{3^n} \cdot 3^1} \cdot \frac{\cancel{3^n}}{\ln n} \quad \frac{\infty}{\infty}$$

$$CF(x) = \frac{\ln(x+1)}{3 \ln x}$$

$$\lim_{x \rightarrow \infty} \frac{\ln(x+1)}{3 \ln x}$$

L'Hop

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x+1}}{\frac{3}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{3(x+1)} = \lim_{x \rightarrow \infty} \frac{\cancel{x}}{3 \cancel{x} (1 + \frac{1}{x})} = \frac{1}{3(1+0)} = \frac{1}{3} < 1$$

$\therefore \sum_{n=3}^{\infty} b_n$ is convergent (Ratio Test)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)



Nice question !!



A series is defined by $a_1 = 1$

$$a_{n+1} = \frac{2 + \sin n}{n^2} a_n$$

Determine whether $\sum a_n$ converges or diverges

$$a_{n+1} = \frac{2 + \sin n}{n^2} a_n$$

$\div a_n$

$$\frac{a_{n+1}}{a_n} = \frac{2 + \sin n}{n^2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2 + \sin n}{n^2} \right|$$

$$\begin{array}{c} -1 \leq \sin n \leq 1 \\ +2 \\ 1 \leq 2 + \sin n \leq 3 \\ \div n^2 \end{array}$$

$$\frac{1}{n^2} \leq \frac{2 + \sin n}{n^2} \leq \frac{3}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} = \frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{3}{n^2} = \frac{3}{\infty} = 0$$

$\therefore$ By Sandwich Thm

$$\lim_{n \rightarrow \infty} \frac{2 + \sin n}{n^2} = 0$$

$\Rightarrow \therefore \sum a_n$ is convergent (Ratio test) ∇



Root Test

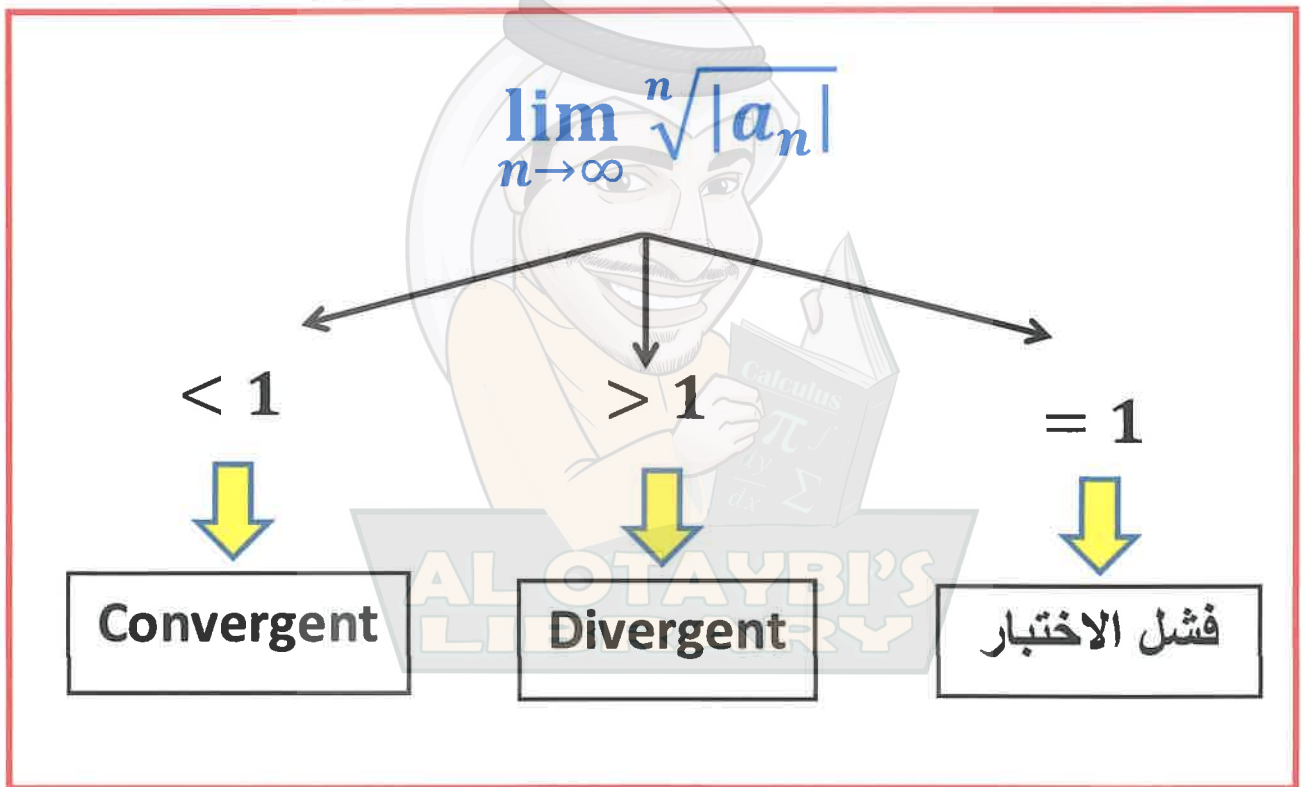


شكله

عند وجود أس
 $n, n^2, \dots$



خطوات الحل



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} e^{-n} \left(1 + \frac{3}{n}\right)^n$$

$$a_n = e^{-n} \left(1 + \frac{3}{n}\right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{e^{-n} \left(1 + \frac{3}{n}\right)^n}$$

$$\lim_{n \rightarrow \infty} \left[e^{-n} \left(1 + \frac{3}{n}\right)^n \right]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} (e^{-n})^{\frac{1}{n}} \left[\left(1 + \frac{3}{n}\right)^n \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} e^{-1} \left(1 + \frac{3}{n}\right) = e^{-1} (1+0) = \frac{1}{e} < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} e^{-n} \left(1 + \frac{3}{n}\right)^n$ is convergent (Root test)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{n^{2n}}{(1+2n^2)^n}$$

$$a_n = \frac{n^{2n}}{(1+2n^2)^n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^{2n}}{(1+2n^2)^n}}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n^{2n}}{(1+2n^2)^n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n^2}{1+2n^2}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^2}}{\cancel{n^2} \left[\frac{1}{n^2} + 2 \right]} = \frac{1}{0+2} = \frac{1}{2} < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{n^{2n}}{(1+2n^2)^n}$ is convergent (Root test)

Determine whether the series is Convergent or divergent.

$$\sum_{n=2}^{\infty} \frac{n^2+1}{(\ln n)^n} \rightarrow \sum a_n$$

$$\lim_{n \rightarrow \infty} [a_n]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left[\frac{n^2+1}{(\ln n)^n} \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{(n^2+1)^{\frac{1}{n}}}{\ln n}$$

$$\frac{\infty}{\infty} \rightarrow \text{L'Hôpital}$$

$$\lim_{x \rightarrow \infty} (x^2+1)^{\frac{1}{x}}$$

$$y = (x^2+1)^{\frac{1}{x}}$$

$$\ln y = \ln (x^2+1)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln (x^2+1)$$

$$\lim_{x \rightarrow \infty} \frac{\ln (x^2+1)}{x}$$

L'Hôpital

$$\lim_{x \rightarrow \infty} \frac{2x}{x^2+1}$$

$$\lim_{x \rightarrow \infty} \frac{2x}{x^2+1} = \lim_{x \rightarrow \infty} \frac{2x}{x^2(1+\frac{1}{x^2})} = \frac{1}{\infty} = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} y = e^0 = 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{(n^2+1)^{\frac{1}{n}}}{\ln n} = \frac{1}{\infty} = 0$$

$\sum a_n$ is convergent (Root test)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-2} \right)^n \cos 2n \rightarrow \sum a_n$$

$$-1 \leq \cos 2n \leq 1$$

$$* \left(\frac{2n+1}{3n-2} \right)^n$$

$$- \left(\frac{2n+1}{3n-2} \right)^n \leq \left(\frac{2n+1}{3n-2} \right)^n \cos 2n \leq \left(\frac{2n+1}{3n-2} \right)^n \rightarrow b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-2} \right)^n$$

$$\lim_{n \rightarrow \infty} [|b_n|]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left[\left| \left(\frac{2n+1}{3n-2} \right)^n \right| \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{n(2 + \frac{1}{n})}{n(3 - \frac{2}{n})} \right| = \left| \frac{2 + \frac{1}{\infty}}{3 - \frac{2}{\infty}} \right| = \frac{2}{3} < 1$$

$\therefore \sum_{n=1}^{\infty} b_n$ is convergent ((Root test))

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent ((B.C.T))

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \left(1 - \frac{\ln 3}{n}\right)^{n^2} \cos 2n \rightarrow \sum_{n=1}^{\infty} a_n$$

$$-1 \leq \cos 2n \leq 1$$

$$* \left(1 - \frac{\ln 3}{n}\right)^{n^2}$$

$$-\left(1 - \frac{\ln 3}{n}\right)^{n^2} \leq \left(1 - \frac{\ln 3}{n}\right)^{n^2} \cos 2n \leq \left(1 - \frac{\ln 3}{n}\right)^{n^2} \quad \rightarrow b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(1 - \frac{\ln 3}{n}\right)^{n^2}$$

$$\lim_{n \rightarrow \infty} [b_n]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{\ln 3}{n}\right)^{n^2}\right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left[1 - \frac{\ln 3}{n}\right]^n = e^{-\ln 3} = e^{\ln 3^{-1}} = \frac{1}{3} < 1$$

$\therefore \sum_{n=1}^{\infty} b_n$ is convergent (Root test)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\ln \left[1 + \frac{1}{2} \sin n \right]}{\left[3 + \frac{1}{n} \right]^n} \rightarrow \sum_{n=1}^{\infty} a_n$$

* $\frac{1}{2}$ $-1 \leq \sin n \leq 1$

$-\frac{1}{2} \leq \frac{1}{2} \sin n \leq \frac{1}{2}$

+1

$\frac{1}{2} \leq 1 + \frac{1}{2} \sin n \leq \frac{3}{2}$

take $\ln$

$\ln \frac{1}{2} \leq \ln \left(1 + \frac{1}{2} \sin n \right) \leq \ln \frac{3}{2}$

$\div \left(3 + \frac{1}{n} \right)^n$

$\frac{\ln \frac{1}{2}}{\left(3 + \frac{1}{n} \right)^n} \leq \frac{\ln \left(1 + \frac{1}{2} \sin n \right)}{\left(3 + \frac{1}{n} \right)^n} \leq \frac{\ln \frac{3}{2}}{\left(3 + \frac{1}{n} \right)^n}$

$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{\ln \frac{3}{2}}{\left(3 + \frac{1}{n} \right)^n}$

$\lim_{n \rightarrow \infty} [b_n]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left[\frac{\ln \frac{3}{2}}{\left(3 + \frac{1}{n} \right)^n} \right]^{\frac{1}{n}}$

$= \lim_{n \rightarrow \infty} \frac{(\ln \frac{3}{2})^{\frac{1}{n}}}{3 + \frac{1}{n}} = \frac{(\ln \frac{3}{2})^0}{3+0} = \frac{1}{3} < 1$

$\therefore \sum_{n=1}^{\infty} b_n$ is convergent (Root test)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)



Nice question



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \left(\frac{3^n + n}{4^n + \sin(n)} \right)^n$$

$$a_n = \left(\frac{3^n + n}{4^n + \sin(n)} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{3^n + n}{4^n + \sin(n)} \right)^n \right|} = \lim_{n \rightarrow \infty} \left[\left| \left(\frac{3^n + n}{4^n + \sin(n)} \right)^n \right| \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{3^n + n}{4^n + \sin(n)}$$

$$\div 4^n$$

$$\lim_{n \rightarrow \infty}$$

$$\frac{\left(\frac{3}{4}\right)^n + \frac{n}{4^n}}{1 + \frac{\sin(n)}{4^n}}$$

$$\frac{0+0}{1+0} = 0 < 1$$

$$\text{Let } f(x) = \frac{x}{4^x}$$

$$\lim_{x \rightarrow \infty} \frac{x}{4^x} \quad \frac{\infty}{\infty}$$

$$\text{L'Hop} \quad \lim_{x \rightarrow \infty} \frac{1}{4^x \ln 4} = \frac{1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{\sin(n)}{4^n}$$

$$-1 \leq \sin(n) \leq 1$$

$$\div 4^n \quad \frac{-1}{4^n} \leq \frac{\sin(n)}{4^n} \leq \frac{1}{4^n}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{4^n} = 0 \quad \lim_{n \rightarrow \infty} \frac{1}{4^n} = 0$$

$\therefore$ By Sandwich Thm

$$\lim_{n \rightarrow \infty} \frac{\sin(n)}{4^n} = 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Root test)



Absolutely Convergent (A.C)
Conditionally Convergent (C.C)
Divergent (D)



خطوات الحل



1) $a_n \rightarrow |a_n|$

2) $\sum |a_n|$

Divergent

Convergent

3) $\lim_{n \rightarrow \infty} |a_n|$

$= 0$

$\neq 0$

4) $|a_n|$ is decreasing

$f'(x) < 0$

$|a_{n+1}| - |a_n| < 0$

$\therefore \sum a_n$

is Conditionally Convergent

$\therefore \sum a_n$

is Absolutely Convergent

$\therefore \sum a_n$

is Divergent

Determine whether the following series is absolutely convergent, conditionally convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$|a_n| = \left| \frac{\tan\left(\frac{1}{n}\right)}{n+1} \right| = \frac{\tan\left(\frac{1}{n}\right)}{n+1}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1}$$

$$\text{Let } b_n = \frac{\frac{1}{n}}{n} = \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad p=2 > 1$$

is convergent (P-series)

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{n+1} \cdot n^2$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{n\left(1 + \frac{1}{n}\right)} \cdot n^2 = \lim_{n \rightarrow \infty} \frac{n \tan\left(\frac{1}{n}\right)}{1 + \frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{n \tan\left(\frac{1}{n}\right)}{1 + \frac{1}{n}} = \frac{1}{1+0} = 1 > 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is convergent (L.C.T)

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is Absolutely Convergent

Determine whether the following series is absolutely convergent, conditionally convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^n \frac{n! 2^n}{(2n)!} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$|a_n| = \left| (-1)^n \frac{n! 2^n}{(2n)!} \right| = \frac{n! 2^n}{(2n)!}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{n! 2^n}{(2n)!}$$

$$|a_n| = \frac{n! 2^n}{(2n)!} \rightarrow |a_{n+1}| = \frac{(n+1)! 2^{n+1}}{(2(n+1))!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)! 2^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{n! 2^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1) \cancel{n!} \cancel{2^n} \cdot 2}{(2n+2)(2n+1) \cancel{(2n)!}} \cdot \frac{\cancel{(2n)!}}{\cancel{n!} \cancel{2^n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(1 + \frac{1}{n}) 2}{(2 + \frac{2}{n}) n (2 + \frac{1}{n})} \right| = \left| \frac{(1+0) 2}{(2+0) \infty (2+0)} \right|$$

$$= \left| \frac{2}{\infty} \right| = 0 < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is convergent (Ratio test)

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is Absolutely Convergent

Determine whether the following series is absolutely convergent, conditionally convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n+1} - \sqrt{n}}{n} \quad \rightarrow a_n$$

Dec 2017

$$|a_n| = \left| (-1)^n \frac{\sqrt{n+1} - \sqrt{n}}{n} \right| = \frac{\sqrt{n+1} - \sqrt{n}}{n}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$$

$$\sum_{n=1}^{\infty} \frac{n+1 - n}{n [\sqrt{n+1} + \sqrt{n}]}$$

$$\text{Let } b_n = \frac{1}{n\sqrt{n}} = \frac{1}{n^{\frac{3}{2}}}$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$

$$p = \frac{3}{2} > 1$$

Convergent
(p-series)

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n [\sqrt{n+1} + \sqrt{n}]} \cdot n^{\frac{3}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n [\sqrt{n(1+\frac{1}{n})} + \sqrt{n}]} \cdot n^{\frac{3}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n [\sqrt{n} \sqrt{1+\frac{1}{n}} + \sqrt{n}]} \cdot n^{\frac{3}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n \sqrt{n} [\sqrt{1+\frac{1}{n}} + 1]} \cdot n^{\frac{3}{2}} = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2} > 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent (L.C.T)

Determine whether the following series is absolutely convergent, conditionally convergent or divergent.

$$\sum_{n=1}^{\infty} (-1)^n \sqrt[n]{e} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$a_n = (-1)^n \sqrt[n]{e} \implies |a_n| = |(-1)^n (e)^{1/n}| = e^{1/n}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} e^{1/n}$$

$$\lim_{n \rightarrow \infty} e^{1/n} = e^{1/\infty} = e^0 = 1 \neq 0$$

$\therefore \sum_{n=1}^{\infty} |a_n|$ is divergent

(N.T.T test of divergence)

$\implies \therefore \sum_{n=1}^{\infty} a_n$ is divergent

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Determine whether the series is absolutely convergent, conditionally convergent or divergent

$$\sum_{n=1}^{\infty} (-1)^n \frac{\ln n + \ln 2}{\ln(n+2)} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$|a_n| = \left| (-1)^n \frac{\ln n + \ln 2}{\ln(n+2)} \right| = \frac{\ln n + \ln 2}{\ln(n+2)}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{\ln n + \ln 2}{\ln(n+2)}$$

~~G.T.N.P.T~~
N.A.B.L
R.B.

$$\lim_{n \rightarrow \infty} \frac{\ln n + \ln 2}{\ln(n+2)} \quad \frac{\infty}{\infty}$$

Let $f(x) = \frac{\ln x + \ln 2}{\ln(x+2)}$

$$\lim_{x \rightarrow \infty} \frac{\ln x + \ln 2}{\ln(x+2)}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{x+2}}$$

$$\lim_{x \rightarrow \infty} \frac{x+2}{x} = \lim_{x \rightarrow \infty} \frac{x(1 + \frac{2}{x})}{x} = 1 + 0 = 1 \neq 0$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is divergent (Test of divergence) $= 1 \neq 0$
N.T.T

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent

Determine whether the series

Dec 2016

$$\sum_{n=1}^{\infty} (-1)^n \tan\left(\frac{1}{n}\right) \rightarrow a_n$$

is absolutely convergent, conditionally convergent or divergent.

$$|a_n| = |(-1)^n \tan\left(\frac{1}{n}\right)| = \left|\tan\left(\frac{1}{n}\right)\right|$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$$

$$\text{Let } b_n = \frac{1}{n}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ divergent (Harmonic)}$$

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n}$$

$$\lim_{n \rightarrow \infty} \tan\left(\frac{1}{n}\right) \neq n$$

$$\lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}} = 1 > 0$$

$$\therefore \sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right) \text{ divergent (L.C.T)}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \tan\left(\frac{1}{n}\right) = \tan 0 = 0$$

$$\text{Let } f(x) = \tan\left(\frac{1}{x}\right)$$

$$f'(x) = \sec^2\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$\therefore \sum_{n=1}^{\infty} a_n$ is conditionally convergent

Determine whether the series

Spring 2019

$$\sum_{n=1}^{\infty} \sin(4n) \left(1 - \frac{1}{3n}\right)^{n^2} \rightarrow a_n$$

absolutely converges, conditionally converges or diverges.

$$|a_n| = \left| \sin(4n) \left(1 - \frac{1}{3n}\right)^{n^2} \right|$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |\sin(4n)| \left(1 - \frac{1}{3n}\right)^{n^2}$$

$$0 \leq |\sin(4n)| \leq 1$$
$$\times \left(1 - \frac{1}{3n}\right)^{n^2}$$

$$0 \leq |\sin(4n)| \left(1 - \frac{1}{3n}\right)^{n^2} \leq \left(1 - \frac{1}{3n}\right)^{n^2} \rightarrow b_n$$

$$\lim_{n \rightarrow \infty} (b_n)^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{3n}\right)^{n^2} \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{3n}\right)^n = e^{-\frac{1}{3}} = \frac{1}{e^{\frac{1}{3}}}$$

$$\therefore \sum_{n=1}^{\infty} |a_n|$$

is convergent (B.C.T)

$\therefore \sum_{n=1}^{\infty} b_n$ is Convergent
(Root test)

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent

Determine whether the series

May 2017

$$\sum_{n=1}^{\infty} \frac{\sin(n!)}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \rightarrow a_n$$

is absolutely convergent, conditionally convergent or divergent.

$$|a_n| = \left| \frac{\sin(n!)}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \right|$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |\sin(n!)| \left| \frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \right|$$

$$0 \leq |\sin(n!)| \leq 1$$

$$\leq \frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2}$$

$$0 \leq |\sin(n!)| \frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \leq \frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \rightarrow b_n$$

$$\frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2}$$

$$\lim_{n \rightarrow \infty} (b_n)^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \left[\frac{1}{5^n} \left(\frac{n+1}{n} \right)^{n^2} \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{1}{5} \left(\frac{n+1}{n} \right)^n$$

$$\lim_{n \rightarrow \infty} \frac{1}{5} \left(1 + \frac{1}{n} \right)^n = \frac{1}{5} e < 1$$

$\therefore \sum_{n=1}^{\infty} |a_n|$ is convergent (R.O.T. Test)

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent

Determine if the series

Spring 2019

$$\sum_{n=1}^{\infty} (-1)^n \frac{\sqrt{n}}{2n+1} \rightarrow a_n$$

absolutely convergent, conditionally convergent or divergent.

$$|a_n| = |(-1)^n \frac{\sqrt{n}}{2n+1}| = \frac{\sqrt{n}}{2n+1}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{\sqrt{n}}{2n+1}$$

$$\text{Let } b_n = \frac{n^{\frac{1}{2}}}{n} = \frac{1}{n^{\frac{1}{2}}}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad p = \frac{1}{2} < 1 \quad \text{divergent (P-series)}$$

$$\lim_{n \rightarrow \infty} \frac{|a_n|}{b_n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{2n+1} \cdot n^{\frac{1}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n(2 + \frac{1}{n})} \cdot n^{\frac{1}{2}} = \frac{1}{2+0} = \frac{1}{2} \neq 0$$

$$\therefore \sum_{n=1}^{\infty} |a_n| \text{ is divergent (L.C.T)}$$

$$\text{Let } f(x) = \frac{\sqrt{x}}{2x+1}$$

$$f'(x) = \frac{(2x+1) \frac{1}{2\sqrt{x}} - \sqrt{x}(2)}{(2x+1)^2} = \frac{\frac{2x+1}{2\sqrt{x}} - 2\sqrt{x}}{(2x+1)^2}$$

$$= \frac{\frac{2x+1-4x}{2\sqrt{x}}}{(2x+1)^2} = \frac{1-2x}{(2x+1)^2 2\sqrt{x}} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$\therefore \sum_{n=1}^{\infty} a_n$ is conditionally convergent.

Determine whether the following series is absolutely convergent, conditionally convergent or divergent.

$$\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^{\frac{3}{4}}} \longrightarrow \sum_{n=1}^{\infty} a_n$$

$$a_n = \frac{\cos(n\pi)}{n^{\frac{3}{4}}} = \frac{(-1)^n}{n^{\frac{3}{4}}}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n^{\frac{3}{4}}} \right|$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{4}}}$$

$$p = \frac{3}{4} < 1$$

is divergent (p-series)

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{1}{n^{\frac{3}{4}}} = \frac{1}{\infty} = 0$$

$$\text{Let } f(x) = \frac{1}{x^{\frac{3}{4}}} = x^{-\frac{3}{4}}$$

$$f'(x) = -\frac{3}{4} x^{-\frac{7}{4}} = \frac{-3}{x^{\frac{7}{4}}}$$

$$< 0 \quad \forall x \in [1, \infty)$$

$\Rightarrow \therefore f(x)$ is decreasing

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is conditionally Convergent (C.C)



Text book question



Let $\{b_n\}$ be a sequence of positive numbers that converges to $\frac{1}{2}$.

Determine whether the given series is absolutely convergent.

$$\sum_{n=1}^{\infty} \frac{b_n^n \cos n\pi}{n} \rightarrow a_n$$

$\therefore b_n$ is converges to $\frac{1}{2} \rightarrow \therefore \lim_{n \rightarrow \infty} b_n = \frac{1}{2}$

$$\sum_{n=1}^{\infty} \frac{(b_n)^n (-1)^n}{n}$$

$$|a_n| = \left| \frac{(b_n)^n (-1)^n}{n} \right| = \frac{(b_n)^n}{n}$$

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{(b_n)^n}{n}$$

$$\lim_{n \rightarrow \infty} \left[\frac{(b_n)^n}{n} \right]^{\frac{1}{n}}$$

$$\lim_{n \rightarrow \infty} \frac{b_n}{n^{\frac{1}{n}}} = \frac{\frac{1}{2}}{1} = \frac{1}{2} < 1$$

$$y = x^{\frac{1}{x}}$$

$$\ln y = \ln x^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \ln x$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$\text{L.H} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \frac{1}{\infty} = 0$$

$$\therefore \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = e^0 = 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is converge
(Root test)

$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n$ is
absolutely convergent



Tricky question !!



Given that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent

Show that $\sum_{n=1}^{\infty} (a_n)^{2n}$ is convergent

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent

$\Rightarrow \therefore \sum_{n=1}^{\infty} |a_n|$ is convergent

$\Rightarrow \therefore \lim_{n \rightarrow \infty} |a_n| = 0$

$$\sum_{n=1}^{\infty} (a_n)^{2n}$$

G.T.H.P.L

N.a.B.L

R.R

$$\lim_{n \rightarrow \infty} \sqrt[n]{(a_n)^{2n}}$$

$$\lim_{n \rightarrow \infty} [(a_n)^{2n}]^{\frac{1}{n}}$$

الأس $\sqrt[n]{}$
Root

$$\lim_{n \rightarrow \infty} (a_n)^2 = 0 < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} (a_n)^{2n}$ is convergent (Root test)

Suppose that $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

Show that $\sum_{n=1}^{\infty} a_n^n$ is absolutely convergent.

Dec 2017

$\therefore \sum_{n=1}^{\infty} a_n$ is absolutely convergent

$\therefore \sum_{n=1}^{\infty} |a_n|$ is convergent

$\therefore \lim_{n \rightarrow \infty} |a_n| = 0$

$\sum_{n=1}^{\infty} a_n^n$

$\sum_{n=1}^{\infty} |a_n^n|$

$\lim_{n \rightarrow \infty} |a_n^n|$

$\lim_{n \rightarrow \infty} |a_n| = 0$

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$\therefore \sum_{n=1}^{\infty} |a_n^n|$ is convergent (Root test)

$\therefore \sum_{n=1}^{\infty} a_n^n$ is absolutely convergent

For what value of p does the following series is absolutely convergent

$$\sum_{n=1}^{\infty} \frac{n^n}{n! e^{np}} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$|a_n| = \frac{n^n}{n! e^{np}} \rightarrow a_{n+1} = \frac{(n+1)^{n+1}}{(n+1)! e^{(n+1)p}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)^n \cdot (n+1)}{(n+1)! e^{np+p}} \cdot \frac{n! e^{np}}{n^n} \right|$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^n (n+1)}{(n+1) n! e^{np} e^p} \cdot \frac{n! e^{np}}{n^n}$$

$$\lim_{n \rightarrow \infty} \frac{(n+1)^n}{n^n e^p} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n \frac{1}{e^p}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \frac{1}{e^p} = e \cdot \frac{1}{e^p} = e^{1-p}$$

$\sum_{n=1}^{\infty} a_n$ is absolutely convergent if

$$e^{1-p} < 1$$

$$1-p < 0$$

$$1-p < 0$$

$$1 < p \Rightarrow p \in (1, \infty)$$

Determine whether the following series is convergent or divergent

$$\frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \dots$$

$$\sum_{n=1}^{\infty} \frac{n^n}{(n+1)^{n+1}} = \sum_{n=1}^{\infty} \frac{n^n}{(n+1)^n \cdot (n+1)}$$

Let $b_n = \frac{n^n}{n^n \cdot n} = \frac{1}{n}$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \Rightarrow \text{divergent} \quad (\text{Harmonic})$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n (n+1)} \cdot \frac{n}{1}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n \cdot \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^{-n} \cdot \frac{n}{n(1+\frac{1}{n})}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{-n} \cdot \frac{1}{1+\frac{1}{n}} = e^{-1} \cdot \frac{1}{1+0} = e^{-1} \neq 0$$

$\therefore \sum_{n=1}^{\infty} b_n$ is divergent

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent ((L.C.T))



11.8 Power series



Find the interval of convergence

Find the radius of convergence

$$1) u_n \rightarrow u_{n+1}$$

$$2) \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right|$$

$$= \infty$$

Interval of convergence

يتم مساواتها بالصفر

Radius of convergence

$$R = 0$$

= |ناتج|



$$= 0$$

Interval of convergence

$(-\infty, \infty)$

Radius of convergence

$$R = \infty$$

Interval of convergence

$$| \text{ناتج} | < 1$$

$$-1 < \text{ناتج} < 1$$

$$a < x < b$$

ثم يتم دراسة الأطراف لمعرفة
سلوكها (مبطللة-مسكرة)

Radius of convergence

$$R = \frac{b - a}{2}$$

Find the interval and radius of convergence of

$$\sum_{n=1}^{\infty} \frac{(x+1)^{2n}}{n 4^n}$$

$$U_n = \frac{(x+1)^{2n}}{n 4^n} \longrightarrow U_{n+1} = \frac{(x+1)^{2(n+1)}}{(n+1) 4^{n+1}} = \frac{(x+1)^{2n+2}}{(n+1) 4^n \cdot 4}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x+1)^{2n} \cdot (x+1)^2}{(n+1) 4^n 4} \cdot \frac{n 4^n}{(x+1)^{2n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n}{(n+1)} \cdot \frac{(x+1)^2}{4} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n}{n(1+\frac{1}{n})} \cdot \frac{(x+1)^2}{4} \right| = \left| \frac{1}{1+0} \cdot \frac{(x+1)^2}{4} \right| = \frac{(x+1)^2}{4}$$

the series is convergent if

$$\frac{(x+1)^2}{4} < 1$$

$$(x+1)^2 < 4$$

$$-2 < x+1 < 2$$

$$-3 < x < 1$$

at $x = -3$

$$\sum_{n=1}^{\infty} \frac{(-2)^{2n}}{n 4^n} = \sum_{n=1}^{\infty} \frac{4^n}{n 4^n} \rightarrow \text{Divergent (Harmonic)}$$

at $x = 1$

$$\sum_{n=1}^{\infty} \frac{2^{2n}}{n 4^n} = \sum_{n=1}^{\infty} \frac{4^n}{n 4^n} \rightarrow \text{Divergent (Harmonic)}$$

• interval of convergent $-3 < x < 1 \Rightarrow (-3, 1)$

(b) Radius of convergence

$$R = \frac{1 - (-3)}{2} = \frac{4}{2} = 2$$

(a) Find the interval of convergence of

$$\sum_{n=1}^{\infty} \frac{n+1}{2n+1} \frac{(x-1)^n}{2^n}$$

(b) Find the radius of convergence

$$U_n = \frac{n+1}{2n+1} \frac{(x-1)^n}{2^n} \rightarrow U_{n+1} = \frac{n+1+1}{2(n+1)+1} \frac{(x-1)^{n+1}}{2^{n+1}} \rightarrow (x-1)^n(x-1)$$

$\xrightarrow{2n+2+1}$ $\xrightarrow{2n+3}$ $\xrightarrow{2^n \cdot 2^1}$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n+2}{2n+3} \cdot \frac{(x-1)^n(x-1)}{2^n \cdot 2^1} \cdot \frac{2n+1}{n+1} \cdot \frac{2^n}{(x-1)^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+2)(2n+1)}{(2n+3)(n+1)} \cdot \frac{(x-1)}{2} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n(1+\frac{2}{n}) \cdot n(2+\frac{1}{n})}{n(2+\frac{3}{n}) \cdot n(1+\frac{1}{n})} \cdot \frac{(x-1)}{2} \right| = \left| \frac{(1+0)(2+0)}{(2+0)(1+0)} \cdot \frac{(x-1)}{2} \right|$$

$$= \left| \frac{2}{2} \cdot \frac{x-1}{2} \right| = \left| \frac{x-1}{2} \right|$$

the series is convergent if

$$\left| \frac{x-1}{2} \right| < 1$$

$$-1 < \frac{x-1}{2} < 1$$

$$-2 < x-1 < 2$$

$$-1 < x < 3$$

at $x = -1$

$$\sum_{n=1}^{\infty} \frac{n+1}{2n+1} \frac{(-2)^n}{2^n} = \sum_{n=1}^{\infty} \frac{n+1}{2n+1} \frac{(-1)^n 2^n}{2^n}$$

$$\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{2n+1}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{n+1}{2n+1} = \lim_{n \rightarrow \infty} \frac{n(1+\frac{1}{n})}{n(2+\frac{1}{n})} = \frac{1+0}{2+0} = \frac{1}{2} \neq 0$$

$\Rightarrow \sum_{n=1}^{\infty} a_n$ is Divergent ((test of divergence))

at $x=3$

$$\sum_{n=1}^{\infty} \frac{n+1}{2n+1} \frac{\cancel{2^n}}{\cancel{2^n}} = \sum_{n=1}^{\infty} \frac{n+1}{2n+1}$$

$$\lim_{n \rightarrow \infty} \frac{n+1}{2n+1} = \lim_{n \rightarrow \infty} \frac{\cancel{n}(1+\frac{1}{n})}{\cancel{n}(2+\frac{1}{n})}$$

$$= \frac{1+0}{2+0} = \frac{1}{2} \neq 0$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (N.T.T) (test of divergence)

$\therefore$ interval of convergent

$$-1 < x < 3 \rightarrow (-1, 3)$$

(b) Find the radius of convergent.

$$\begin{aligned} \text{radius of convergent } r &= \frac{b-a}{2} \\ &= \frac{3-(-1)}{2} \\ &= \frac{4}{2} = 2 \end{aligned}$$

$$\Rightarrow R=2$$

(a) Find the interval

of convergence of

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x-3)^n}{n 2^n}$$

(b) Find the radius of convergence

$$u_n = \frac{(-1)^n (x-3)^n}{n 2^n} \rightarrow u_{n+1} = \frac{(-1)^{n+1} (x-3)^{n+1}}{(n+1) 2^{n+1}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1} (x-3)}{(n+1) 2^{n+1}} \cdot \frac{n 2^n}{(x-3)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \cdot \frac{x-3}{2} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n(1+\frac{1}{n})} \cdot \frac{x-3}{2} \right| = \left| \frac{x-3}{2} \right|$$

the series is convergent when

$$\left| \frac{x-3}{2} \right| < 1$$

$$-1 < \frac{x-3}{2} < 1$$

$$-2 < x-3 < 2$$

$$1 < x < 5$$

$$R = \frac{5-1}{2} = \frac{4}{2} = 2 \rightarrow \therefore \text{Radius of conv} = 2$$

at $x=1$

$$\sum_{n=1}^{\infty} \frac{(-1)^n (-2)^n}{n 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^n 2^n}{n 2^n}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n} \rightarrow \text{Divergent (Harmonic)}$$

$$\text{at } x=5 \quad \sum_{n=1}^{\infty} \frac{(-1)^n (2)^n}{n 2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$$

$$\text{Let } f(x) = \frac{1}{x} \rightarrow f'(x) = -\frac{1}{x^2} < 0 \quad \forall x \in (1, 5)$$

$f(x)$ is decreasing

$$\because \sum_{n=1}^{\infty} |a_n| \text{ is decreasing} \Rightarrow \because \sum_{n=1}^{\infty} a_n \text{ is convergent.}$$

$\therefore$ the interval of convergence is $[1, 5]$ (A.S.T)



Text book question



Find the radius and interval of convergence of the power series

$$\sum_{n=2}^{\infty} \frac{(x+2)^n}{2^n \ln n}$$

$$U_n = \frac{(x+2)^n}{2^n \ln n} \longrightarrow U_{n+1} = \frac{(x+2)^{n+1}}{2^{n+1} \ln(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \frac{(x+2)^n (x+2)}{2^n 2 \ln(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+2)^n (x+2)}{2^n 2 \ln(n+1)} \cdot \frac{2^n \ln n}{(x+2)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\ln n}{\ln(n+1)} \cdot \frac{x+2}{2} \right| = \left| (1) \frac{x+2}{2} \right| = \left| \frac{x+2}{2} \right|$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\ln(x+1)}$$

L'Hop $\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{x+1}} = \lim_{x \rightarrow \infty} \frac{x+1}{x} = \lim_{x \rightarrow \infty} \frac{x[1+\frac{1}{x}]}{x} = 1+0 = 1$

The series is convergent if

$$\left| \frac{x+2}{2} \right| < 1$$

$$-1 < \frac{x+2}{2} < 1$$

$$-2 < x+2 < 2$$

$$-4 < x < 0$$

$\therefore$ The radius of convergence $R = \frac{0 - (-4)}{2} = 2$

at $x = -4$

$$\sum_{n=2}^{\infty} \frac{(-2)^n}{2^n \ln n} = \sum_{n=2}^{\infty} \frac{(-1)^n 2^n}{2^n \ln n}$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$$

$$\text{Let } b_n = \frac{1}{\ln n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0$$

$$f(x) = \frac{1}{\ln x}$$

$$f'(x) = \frac{0 - (1) \cdot \frac{1}{x}}{(\ln x)^2}$$

$$= \frac{-1}{x (\ln x)^2} < 0 \quad \forall x \in [2, \infty)$$

$\therefore f(x)$ is decreasing on $[2, \infty)$

$\therefore \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ is convergent (A.S.T)

at $x = 0$

$$\sum_{n=2}^{\infty} \frac{2^n}{2^n \ln n} = \sum_{n=2}^{\infty} \frac{1}{\ln n}$$

$$\ln n < n \quad \forall n > 0$$

$$\frac{1}{\ln n} > \frac{1}{n}$$

$\sum_{n=2}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$\therefore \sum_{n=2}^{\infty} \frac{1}{\ln n}$ is divergent (B.C.T)

$\Rightarrow \therefore$ The interval of convergence = $[-4, 0)$

Find the radius of convergence of

Dec 2017

$$\sum_{n=2}^{\infty} (-1)^n \frac{(x-3)^n}{2^n \ln n}$$

$$U_n = (-1)^n \frac{(x-3)^n}{2^n \ln n} \longrightarrow U_{n+1} = (-1)^{n+1} \frac{(x-3)^{n+1}}{2^{n+1} \ln(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1} (x-3)}{2^{n+1} 2 \ln(n+1)} \cdot \frac{2^n \ln n}{(x-3)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\ln n}{\ln(n+1)} \right|$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\ln(x+1)}$$

L'Hop

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{x+1}}$$

$$= \lim_{x \rightarrow \infty} \frac{x+1}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{x} \left[1 + \frac{1}{x} \right] = 1 + 0 = 1$$

The series is convergent if

$$\left| \frac{x-3}{2} \right| < 1$$

$$-1 < \frac{x-3}{2} < 1$$

$$-2 < x-3 < 2$$

$$1 < x < 5$$

$$R = \frac{5-1}{2} = \frac{4}{2} = 2$$

$\therefore$ The radius of convergence = 2

Find the radius of convergence of the series

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$$\sum_{n=2}^{\infty} \frac{\ln n}{n^2} x^{2n}$$

$$U_n = \frac{\ln n}{n^2} x^{2n} \longrightarrow U_{n+1} = \frac{\ln(n+1)}{(n+1)^2} x^{2(n+1)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$= \frac{\ln(n+1)}{(n+1)^2} x^{2n+2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\ln(n+1) x^{2n+2}}{(n+1)^2} \right|$$

$$\times \frac{n^2}{\ln n} \frac{1}{x^{2n}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{\ln(n+1)}{\ln n} \right|$$

$$\frac{n^2}{(n+1)^2} x^2 = \left(\frac{n}{n+1} \right)^2 x^2 = x^2$$

$$\lim_{x \rightarrow \infty} \frac{\ln(x+1)}{\ln x}$$

L'HOP

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x+1}}{\frac{1}{x}}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x+1}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x(1+\frac{1}{x})} = \frac{1}{1+0} = 1$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{(n(1+\frac{1}{n}))^2}$$

$$= \frac{1}{(1+0)^2} = 1$$

∴ The series is convergent if

$$x^2 < 1$$

$$|x| < 1$$

$$-1 < x < 1$$

$$\text{Radius of convergence} \longrightarrow R = \frac{1-(-1)}{2} = 1$$

Find the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x+3)^n}{(2n)!}$$

Dec 2016

$$U_n = \frac{(-1)^n (x+3)^n}{(2n)!} \rightarrow U_{n+1} = \frac{(-1)^{n+1} (x+3)^{n+1}}{(2(n+1))!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\cancel{(x+3)^n} (x+3)^1}{(2n+2)!} \cdot \frac{(2n)!}{\cancel{(x+3)^n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(2n)!}{(2n+2)(2n+1)(2n)!} \cdot \cancel{x+3} \right| = 0$$

∴ The series is convergent on $x \in \mathbb{R}$

∴ The interval of convergence = $(-\infty, \infty)$

The radius of convergence $R = \infty$

Find the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} n! (2x - 1)^n$$

$$U_n = n! (2x - 1)^n \longrightarrow U_{n+1} = (n+1)! (2x - 1)^{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)! (2x-1)^{n+1}}{n! (2x-1)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1) \cancel{n!} (2x-1)^{\cancel{n}} \cdot (2x-1)^1}{\cancel{n!} (2x-1)^{\cancel{n}}} \right| = \infty$$

The series is convergent when $2x - 1 = 0$

$$2x = 1$$

$$x = \frac{1}{2}$$

The radius of convergence $R = 0$

Find the interval of convergence of

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^n}{n+1} (x-4)^n$$

$$U_n = (-1)^n \frac{n^n}{n+1} (x-4)^n \rightarrow U_{n+1} = (-1)^{n+1} \frac{(n+1)^{n+1}}{n+2} (x-4)^{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} \frac{(n+1)^{n+1}}{n+2} (x-4)^{n+1}}{(-1)^n \frac{n^n}{n+1} (x-4)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)^{n+1}}{n^n} \frac{(n+1)^2}{n+2} (x-4) \right|$$

$$\lim_{n \rightarrow \infty} \left| \left(\frac{n+1}{n} \right)^n \frac{[n(1+\frac{1}{n})]^2}{n(1+\frac{2}{n})} (x-4) \right|$$

$$\lim_{n \rightarrow \infty} \left| \left(1 + \frac{1}{n} \right)^n \frac{n(1+\frac{1}{n})^2}{(1+\frac{2}{n})} (x-4) \right| = \left| e \frac{\infty(1+0)^2}{1+0} (x-4) \right| = \infty$$

$\Rightarrow \therefore \sum U_n$ is convergent at $x=4$

$\Rightarrow \text{Radius} = 0$



Tricky question !!



Given a sequence $\{a_n\}$ such that $a_1 = 2$

$$a_{n+1} = \frac{4n + 101}{3n + 2} a_n, \quad n \geq 1$$

(a) Show that $\sum_{n=0}^{\infty} a_n$ is divergent series

(b) Find the radius of convergence of the power series

$$\sum_{n=0}^{\infty} a_n (3x + 2)^n$$

(a)

$$a_{n+1} = \frac{4n+101}{3n+2} a_n$$

$\div a_n$

$$\frac{a_{n+1}}{a_n} = \frac{4n+101}{3n+2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{4n+101}{3n+2} \right| = \lim_{n \rightarrow \infty} \left| \frac{n(4 + \frac{101}{n})}{n(3 + \frac{2}{n})} \right| = \frac{4+0}{3+0} = \frac{4}{3} > 1$$

$\Rightarrow \therefore \sum_{n=0}^{\infty} a_n$ is divergent (Ratio test)

(b)

$$U_n = a_n (3x+2)^n \rightarrow U_{n+1} = a_{n+1} (3x+2)^{n+1}$$

$$\lim_{n \rightarrow \infty} \left| \frac{U_{n+1}}{U_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (3x+2)^{n+1}}{a_n (3x+2)^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} (3x+2) \right| = \left| \frac{4}{3} (3x+2) \right| = \left| 4x + \frac{8}{3} \right|$$

$\therefore$ the series is convergence

$$\left| 4x + \frac{8}{3} \right| < 1$$

$$-1 < 4x + \frac{8}{3} < 1$$

$$-\frac{11}{12} < 4x < -\frac{5}{12}$$

$$\div 4 \quad -\frac{11}{12} < x < -\frac{5}{12}$$

$$R = -\frac{5}{12} - \left(-\frac{11}{12}\right) = \frac{-5+11}{12} = \frac{6}{12} = \frac{1}{2} = \left(\frac{1}{4}\right)$$



Multiple choice



Which of the following is true for the series $\sum_{n=1}^{\infty} a_n$?

(a) If the series is convergent, then it is also absolutely convergent.

(b) If the series is alternating, then it is convergent.

(c) If $\lim_{n \rightarrow \infty} a_n = L$, then $\sum_{n=1}^{\infty} a_n = L$.

(d) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series is convergent.

(e) If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series diverges.

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(e) ✓

(Test of divergence)



Multiple choice



Select the power series whose interval of convergence is $[-1, 1)$

(a) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

(b) $\sum_{n=0}^{\infty} \frac{x^n}{n}$

(c) $\sum_{n=0}^{\infty} \frac{x^n}{n^2}$

(d) $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n}$

(e) $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n^2}$

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(b) ✓

If $\sum_{n=1}^{\infty} C_n 3^n$ is convergent.

Prove that $\sum_{n=1}^{\infty} (-1)^n C_n 2^n$ is convergent.

Sol #1

$$\sum_{n=1}^{\infty} C_n (-2)^n \quad ?$$

$$3^n > (-2)^n \quad \forall n > 1$$

$$3^n > (-2)^n$$

$$C_n 3^n > C_n (-2)^n$$

$\therefore \sum_{n=1}^{\infty} C_n 3^n$ is convergent

$\therefore \sum_{n=1}^{\infty} C_n (-2)^n$ is convergent (B.C.T)

Sol #2

$\therefore \sum_{n=1}^{\infty} C_n 3^n$ is convergent

$\therefore$ The interval of convergence contain $(-3, 3)$

$$\therefore -2 \in (-3, 3)$$

$\therefore \sum_{n=1}^{\infty} C_n (-2)^n$ is convergent

If $\sum_{n=1}^{\infty} (-1)^n C_n$ is absolutely convergent.

Prove that $\sum_{n=1}^{\infty} \frac{C_n}{2^n}$ is absolutely convergent.

Sol #1

$\therefore \sum_{n=1}^{\infty} (-1)^n C_n$ is absolutely convergent

$\therefore \sum_{n=1}^{\infty} |(-1)^n C_n|$ is convergent

$\therefore \sum_{n=1}^{\infty} |C_n|$ is convergent

$$1 > \left(\frac{1}{2}\right)^n \quad \forall n > 1$$

$$C_n > C_n \left(\frac{1}{2}\right)^n$$

$$|C_n| > |C_n| \left(\frac{1}{2}\right)^n$$

$\therefore \sum_{n=1}^{\infty} |C_n|$ is convergent $\rightarrow \therefore |C_n| \left(\frac{1}{2}\right)^n$ is conv. (B.C.T)

$\therefore \sum_{n=1}^{\infty} \frac{|C_n|}{2^n}$ is abs. conv.

Sol #2

$\therefore \sum_{n=1}^{\infty} (-1)^n C_n$ is absolutely convergent

$\therefore \sum_{n=1}^{\infty} |(-1)^n C_n|$ is convergent

$\therefore \sum_{n=1}^{\infty} |C_n|$ is convergent

$\therefore$ The interval of convergence contain $(-1, 1)$

$$\therefore \frac{1}{2} \in (-1, 1)$$

$\therefore \sum_{n=1}^{\infty} \frac{|C_n|}{2^n}$ is convergent

$\therefore \sum_{n=1}^{\infty} \frac{C_n}{2^n}$ is absolutely convergent

Show that if the sequence a_n is bounded, then

$$\sum_{n=1}^{\infty} \frac{a_n}{n^2}$$

is convergent

$\therefore a_n$ is bounded

$$\therefore a \leq a_n \leq b$$

$$\div n^2$$

$$\frac{a}{n^2} \leq \frac{a_n}{n^2} \leq \frac{b}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{b}{n^2}$$

($p = 2 > 1$)

Convergent

(p-series)

$$\therefore \sum_{n=1}^{\infty} \frac{a_n}{n^2}$$

is convergent

(B.C.T)

